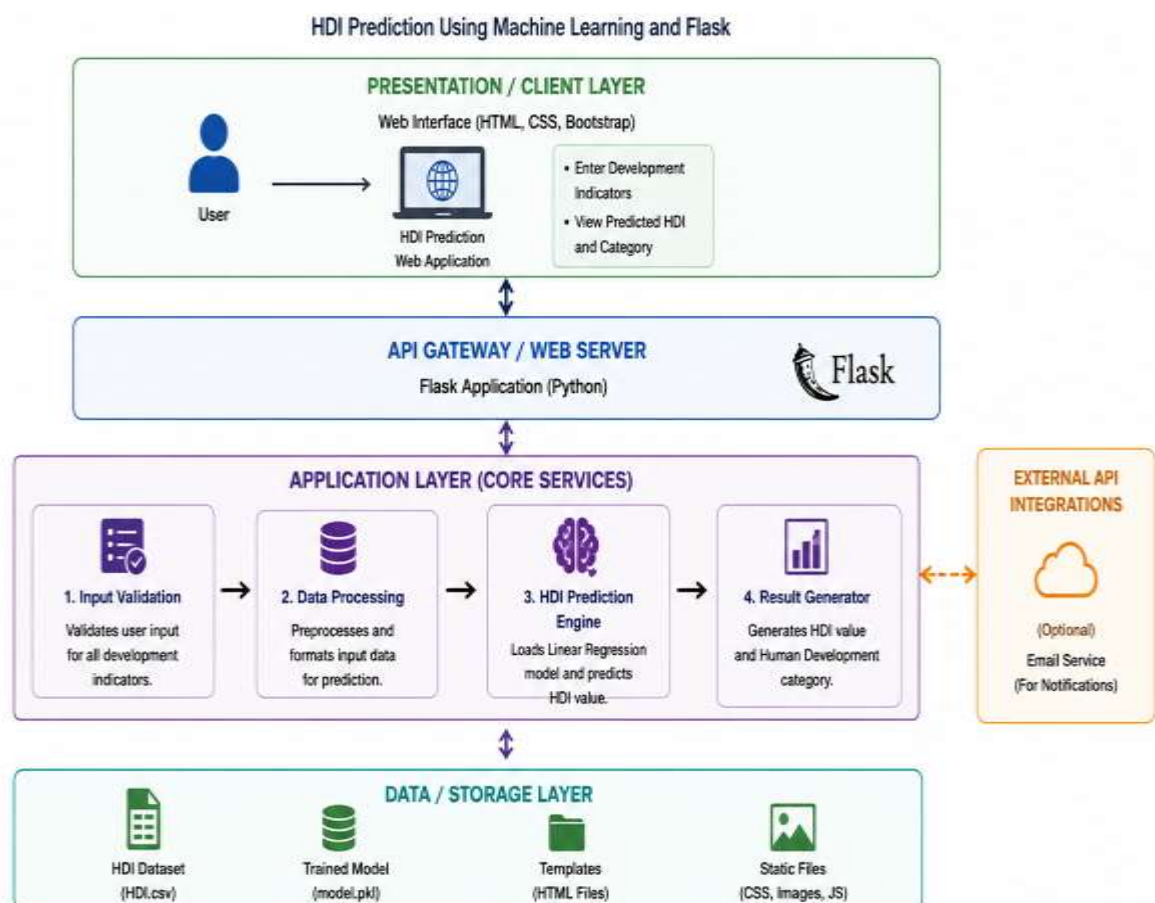


Solution Architecture

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	Provides a user-friendly web interface where users enter development indicator values and view the predicted HDI and Human Development category.	HTML5, CSS3, Bootstrap
<i>[API Gateway]</i>	Receives user requests, processes form data, communicates with the machine learning model, and returns prediction results.	Flask (Python)
<i>[Microservices]</i>	<i>Performs input validation, preprocesses data, loads the trained Linear Regression model, predicts the HDI value, and generates the final output.</i>	<i>Python, Scikit-learn, Pandas, NumPy, Pickle</i>
<i>[Database]</i>	<i>Stores the HDI dataset used for training, the trained machine learning model, HTML templates, CSS files, and other project resources.</i>	<i>CSV Dataset, Pickle (.pkl), Templates, Static Files</i>